# ANOVA Case Study

Background Music & Retail Spending Analysis

## 1. Data Collection (Daily Spending in USD)

A study was conducted to determine if the type of background music influenced the amount spent by customers.

| No Music (G1) | Classical (G2) | Pop (G3) |
|---|---|---|
| 15 | 25 | 20 |
| 20 | 30 | 25 |
| 12 | 22 | 18 |
| 13 | 23 | 17 |
| **Mean: 15** | **Mean: 25** | **Mean: 20** |

**Grand Totals:** N = 12 | Grand Sum = 240 | ***Grand Mean ($\bar{x}_G$) = 20***

## 2. Step-by-Step Calculation

### Step A: Sum of Squares Between ($SS_{bet}$)

Measures variance between the different music groups.

$SS_{bet} = \Sigma n(\bar{x}_i - \bar{x}_G)^2$

1. No Music: $4 \times (15 - 20)^2 = 4 \times 25 = 100$
2. Classical: $4 \times (25 - 20)^2 = 4 \times 25 = 100$

3. Pop: $4 \times (20 - 20)^2 = 4 \times 0 = 0$

**Total $SS_{bet} = 200$**

### Step B: Sum of Squares Within ($SS_{wit}$)

Measures variance within each group (individual differences).

$SS_{wit} = \Sigma (x - \bar{x}_{group})^2$

1. No Music: $(0)^2 + (5)^2 + (-3)^2 + (-2)^2 = 38$
2. Classical: $(0)^2 + (5)^2 + (-3)^2 + (-2)^2 = 38$
3. Pop: $(0)^2 + (5)^2 + (-2)^2 + (-3)^2 = 38$

**Total $SS_{wit} = 114$**

### Step C: Mean Squares (MS) and F-Statistic

$df_{bet} = 3 - 1 = 2$ | $df_{wit} = 12 - 3 = 9$

$MS_{bet} = 200 / 2 = 100$

$MS_{wit} = 114 / 9 = 12.67$

$$F = MS_{bet} / MS_{wit} = 100 / 12.67 = \mathbf{7.89}$$

## 3. Final Conclusion

**Critical Value:** For df(2, 9) at $\alpha = 0.05$, the critical value is **4.26**.

**Decision:** Since the Calculated F (7.89) is greater than the Critical Value (4.26), we **Reject the Null Hypothesis**.

**Result:** The type of music played in the shop has a statistically significant impact on spending. Classical music led to the highest average spending ($25).